NELVARX SYSTEMS AB

NELVARX MATHEMATICAL LABORATORY

GEOGRAPHY OF COMPUTE

A Mathematical Model for Structural AI Data Center Siting Feasibility

PUBLIC RESEARCH DEMONSTRATION CASE

ID	Version	Date	Status
NELVARX-AIDC-GEO-001	v0.2 EN	19.10.2026	Synthetic demonstration

All A–D inputs are synthetic. The model does not prove the suitability of any real site.

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01 Executive summary

This report presents a bounded public mathematical model of an AI data center infrastructure configuration. Four hypothetical alternatives A–D are used. No candidate represents an actual country, city or land parcel.

The revised electricity constraint now depends on IT workload and PUE, rather than an inconsistent generic grid threshold. Under the specified assumptions B and D pass, C fails, and A remains undetermined. These are results obtained from synthetic inputs.

The finding is intentionally narrow: formal models can expose conflicting conditions and conditional trade-offs. They cannot certify real-world grid connections, permits, water resources or engineering readiness.

02 Object, question and boundaries

The object x is a hypothetical combination of a site, an IT workload, cooling, power supply, connection timing and network connectivity. This is not a ranking of nations or geographical regions.

The modelling question is whether mandatory requirements are jointly satisfiable, which alternatives remain unknown, whether a target configuration is reachable and how results change under parameter disturbances.

The scope excludes actual grid connection offers, property rights, full project finance, engineering water withdrawals, design certification, legal assessment and site-level predictions.

03 Structural system and dependencies

Computing workloads create electrical demand and heat. PUE connects IT power with total facility power. Grid capacity and the time needed to secure it introduce distinct mandatory constraints. Tariff contributes to spending but does not repair inadequate capacity.

Climate is retained as an explanatory attribute rather than mechanically converted into PUE. Candidate PUE values are supplied independently. Water stress is a synthetic proxy, not a verified catchment assessment.

The key coupling is that a low price cannot remove a capacity, water or timing restriction.

04 Variables, parameters and units

For each candidate i, define xi = (Gi, Li, Wi, Ni, pi, ηi, Ti). G is confirmed intake capacity in MW; L is connection lead time in months; W is a synthetic water-stress proxy in percent; N is network latency in ms; p is price in EUR/kWh; η is PUE; T is average temperature in degrees C.

Global parameters are P_IT = 80 MW, H = 8760 h/year and electrical reserve μ = 0.05. These are demonstration inputs, not universally applicable industrial standards. PUE is dimensionless and P_IT denotes IT equipment demand rather than whole-facility power.

An unavailable value is represented as UNKNOWN, not zero. Inputs and thresholds are defined assumptions; they are not asserted to come from an engineering code.

x = (G, L, W, N, price, PUE, T)

Parameter	Value	Unit
P_IT	80	MW / МВт
H	8760	hours/year
μ	0.05	fraction
L_max	30	months
W_max	60	%
N_max	150	ms

05 Equations for power, energy and cost

Total facility power is P_facility,i = P_IT × ηi. Power intake including reserve is G_min,i = P_IT × ηi × (1 + μ). The condition Gi ≥ G_min,i is a model-level test of intake sufficiency.

Annual electricity use is Ei = P_facility,i × H, expressed in MWh when power is MW and time is hours. Equivalently, E_GWh,i = P_facility,i × 8.76.

Annual electricity-only spending in million EUR is Ci = E_GWh,i × pi, because 1 GWh is one million kWh. CAPEX, network charges, taxes and financing are omitted.

P_facility = 80 × PUE [MW]

G_min = 80 × PUE × 1.05 [MW]

E_GWh = 80 × PUE × 8.76 [GWh/year]

C_EUR_m = E_GWh × price_EUR_per_kWh

06 Configuration space

Let Ω_raw denote considered combinations of parameters and architectures. Each x is subject to mandatory conditions K = {k_power, k_lead, k_water, k_latency}.

The admissible set is Ω_adm = {x in Ω_raw: every kj(x) is TRUE}. Ω_rej comprises candidates with at least one FALSE mandatory test. Ω_unknown comprises candidates with no established violation but at least one UNKNOWN result.

These sets are definitions within the bounded research model. They do not imply that all actual sites are included in Ω_raw or that real-world suitability is intrinsically binary.

Ω_adm = {x ∈ Ω_raw : all mandatory tests are TRUE}

Ω_rej = {x ∈ Ω_raw : some mandatory test is FALSE}

Ω_unknown = Ω_raw \ (Ω_adm ∪ Ω_rej)

07 Mandatory constraints and unknowns

Four synthetic constraints are imposed: Gi ≥ 80 × PUEi × 1.05; Li ≤ 30 months; Wi ≤ 60%; Ni ≤ 150 ms. The water and latency thresholds are illustrative, not regulatory limits or verified engineering specifications.

The classification procedure is REJECTED if any mandatory test is FALSE; otherwise UNDETERMINED if one or more tests are UNKNOWN; otherwise ADMISSIBLE if all are TRUE.

A TRUE comparison proves only that a supplied input satisfies the stated formula. It does not establish that the input corresponds to an actual measured physical property.

Test	Condition
Power	G ≥ 80 × PUE × 1.05
Lead	L ≤ 30 months
Water	W ≤ 60%
Latency	N ≤ 150 ms

08 Synthetic inputs and admissibility results

The tables disclose all synthetic candidate inputs. For B, total electricity demand is 108 MW and reserved minimum intake is 113.4 MW; the assumed 115 MW supply passes. D requires 105 MW and has 110 MW.

C violates both the connection lead-time limit (35 versus 30 months) and the synthetic water proxy limit (85 versus 60%). A has unknown available intake capacity, so its absence of a recorded violation does not amount to confirmation.

These classifications are relative to the disclosed rules and inputs. They are not geographical findings.

ID	€/kWh	°C	Grid MW	Lead mo	Water %	PUE	ms
A	0.065	8	?	20	45	1.28	100
B	0.115	18	115	18	30	1.35	120
C	0.055	6	110	35	85	1.22	80
D	0.142	13	110	26	25	1.25	90

Case	MW demand	MW required	MW available	Status
A	102.40	107.52	?	Undetermined
B	108.00	113.40	115	Admissible
C	97.60	102.48	110	Rejected
D	100.00	105.00	110	Admissible

09 Energy and tariff calculations

Energy is calculated under continuous operation for 8760 hours. This is a simplifying assumption: actual energy consumption depends on utilization and equipment duty cycles.

C offers the lowest modelled electricity cost, yet fails the mandatory gates. It therefore cannot be selected simply because its price is lower.

The displayed values are rounded for readability; calculations use unrounded intermediate numbers.

Case	PUE	Energy GWh	€/kWh	Cost €m
A	1.28	897.024	0.065	58.30656
B	1.35	946.080	0.115	108.79920
C	1.22	854.976	0.055	47.02368
D	1.25	876.000	0.142	124.39200

10 Mathematical reachability

Feasibility of a target configuration differs from reachability from the current system state. Define graph states S0 (initial), S1 (project approval), S2 (construction and connection) and S3 (operational readiness). Edges represent transitions permitted under scenario conditions.

For a simplified schedule, connection duration L is the sum of non-overlapping stage durations: L = τ_1 + τ_2 + τ_3. For B: 5 + 9 + 4 = 18 months; for D: 8 + 12 + 6 = 26 months. These values are constructed examples, not project plans.

A time-feasible path in the toy model requires authorised transitions and Στj ≤ 30. Actual reachability also requires verified financing, contracts, equipment and permits, which are absent from this study.

Case	Stage 1	Stage 2	Stage 3	Total
B	5	9	4	18
D	8	12	6	26

11 Mathematical stability

Given a specified disturbance set Δ, a candidate is robustly admissible in the model if it passes every mandatory test for every δ in Δ: Robust(x, Δ) if and only if for all δ in Δ, x(δ) belongs to Ω_adm.

Three separate stress scenarios are considered: PUE increases by 0.04; connection is delayed by 4 months; the electricity tariff rises by 40%. Tariff is not a mandatory gate and affects only cost. PUE affects both intake demand and energy.

Under the PUE shock B needs 116.76 MW, exceeding its 115 MW intake, while D needs 108.36 MW, less than its 110 MW capacity. D remains admissible under each of the three separate tests. This does not prove stability under all conceivable disturbances.

Case	Base PUE	PUE +0.04	Need MW	Grid MW	Result
B	1.35	1.39	116.76	115	Rejected
D	1.25	1.29	108.36	110	Admissible

12 Multi-objective optimization

Only members of Ω_adm enter the optimization comparison. The illustrative objective vector minimizes f(x) = (electricity spending, lead time, PUE, water proxy). There is no arbitrary scalar combined score.

Candidate x dominates y if it is no worse on every selected objective and strictly better on at least one. Here B has lower electricity spending and faster connection; D has lower PUE and a lower synthetic water indicator.

Consequently, both B and D are on the synthetic Pareto frontier. Declaring one universal winner would require specified preferences or further mandatory requirements.

Measure	B	D
Electricity €m	108.79920	124.39200
Lead months	18	26
PUE	1.35	1.25
Water proxy %	30	25

13 Sensitivity experiments

Reducing the connection limit from 30 to 24 months keeps B admissible, rejects D, leaves A undetermined, and C rejected.

Increasing B electricity price by 40% changes annual modelled energy spending from EUR 108.7992 million to EUR 152.31888 million. No mandatory admissibility test changes in this tariff scenario.

If the missing intake capacity of A is subsequently assumed to be 110 MW, it passes because 110 ≥ 107.52 MW. If measured capacity is below 107.52 MW, the power constraint fails.

Scenario	B	D	Observation
Lead ≤24	Admissible	Rejected	Deadline sensitive
PUE +0.04	Rejected	Admissible	Capacity sensitive
Tariff +40%	Admissible	Admissible	Cost changes only

14 Risks, unknowns and verification

The largest risks include false precision in inputs, unit mistakes, power/PUE inconsistency, oversimplified water indicators, conflating latency with full network availability, uncertain permitting and incompatible time horizons.

A real project would require a dated grid connection commitment, hourly energy and climate data, a technically specified cooling design, water basin evidence, legal review, verified capital cost and specialists who can interpret each source.

Current verification covers reproducibility of arithmetic, consistent definitions and correct classification of synthetic cases. No independent engineering audit has been undertaken.

Risk / Риск	Consequence / Последствие	Required verification / Проверка
Grid source	False positive	Dated utility evidence
Cooling/PUE	Wrong capacity	Engineering model
Water proxy	False restriction	Basin-specific assessment
Synthetic values	No real-world inference	Primary site data

15 Why a mathematical model is not a proof

A mathematical model is a formal representation of selected object properties under explicit assumptions. It allows logical exploration of consequences, comparative analysis and detection of inconsistencies. It cannot turn its assumptions into observed facts.

One may establish a conditional proposition inside the model, such as: if G ≥ G_min and every other mandatory requirement is satisfied, x is admissible. That is a statement about the formal model, not proof that a real facility has capacity G or that the constraint set is complete.

Real-world claims require measurements, source provenance, applicability review, calibration, validation and domain expertise. Even successful tests of several scenarios do not establish universal validity of the Nelvarx methodology.

16 Conclusion and readiness

The model produces an explicit computational partition of hypothetical alternatives into admissible, rejected and undetermined. The prior inconsistency between intake MW and PUE-dependent total demand has been corrected.

In the baseline B and D pass all four mandatory tests; C violates timing and water constraints; A needs power confirmation. Under the stated stress tests D retains admission while B fails the added PUE shock.

The result is suitable as a public research demonstration and a starting point for a real-data pilot. It does not verify actual siting suitability, a deployed commercial software product or general validity of the underlying methodology.

17 Appendix: reproducibility and provenance

The disclosed computation is: for each i calculate P_facility = 80 × PUE; G_min = P_facility × 1.05; E_GWh = P_facility × 8.76; C_EUR_m = E_GWh × price. Then evaluate four mandatory constraints and preserve UNKNOWN.

The A–D numbers are deliberately constructed synthetic scenarios. No real geographical dataset is used. The proprietary Nelvarx Systems AB methodology is not included in this public report.

No running Nelvarx software-analytical platform is claimed to have performed these calculations. Reproducibility here concerns disclosed computations and inputs rather than operational proprietary software.